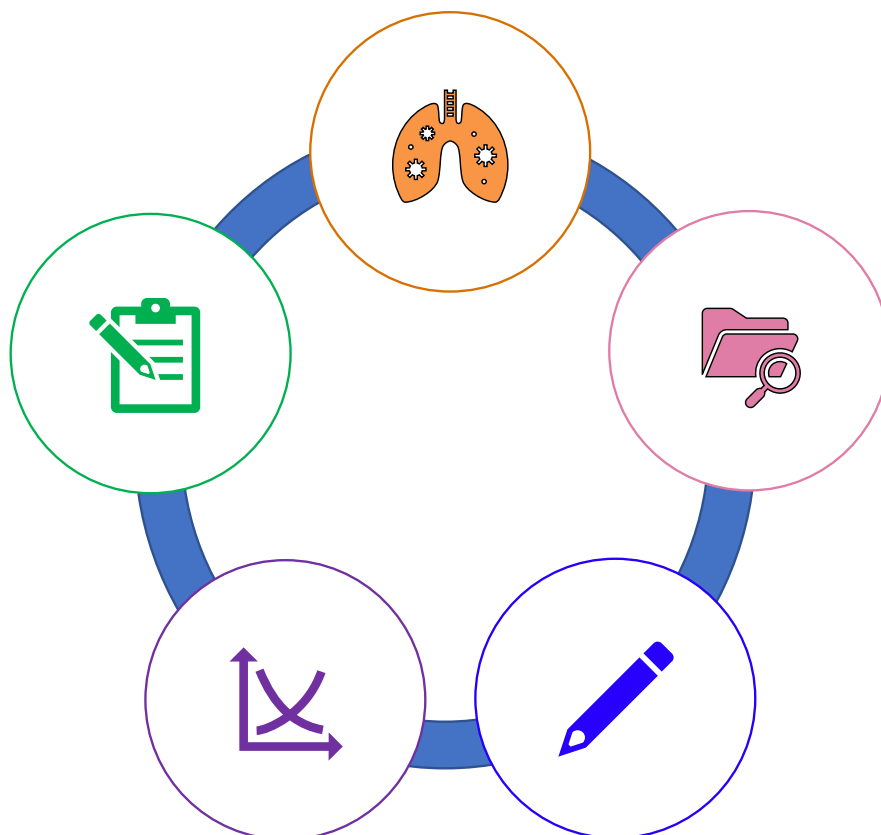


Understanding and Mitigating Bias in Imaging Artificial Intelligence

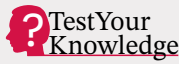
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See the invited commentary by [Rouzrokh and Erickson](#) in this issue.



Artificial intelligence (AI) algorithms are prone to bias at multiple stages of model development, with potential for exacerbating health disparities. However, bias in imaging AI is a complex topic that encompasses multiple coexisting definitions. *Bias* may refer to unequal preference to a person or group owing to preexisting attitudes or beliefs, either intentional or unintentional. However, *cognitive bias* refers to systematic deviation from objective judgment due to reliance on heuristics, and *statistical bias* refers to differences between true and expected values, commonly manifesting as systematic error in model prediction (ie, a model with output unrepresentative of real-world conditions). Clinical decisions informed by biased models may lead to patient harm due to action on inaccurate AI results or exacerbate health inequities due to differing performance among patient populations. However, while inequitable bias can harm patients in this context, a mindful approach leveraging equitable bias can address underrepresentation of minority groups or rare diseases. Radiologists should also be aware of bias after AI deployment such as automation bias, or a tendency to agree with automated decisions despite contrary evidence. Understanding common sources of imaging AI bias and the consequences of using biased models can guide preventive measures to mitigate its impact. Accordingly, the authors focus on sources of bias at stages along the imaging machine learning life cycle, attempting to simplify potentially intimidating technical terminology for general radiologists using AI tools in practice or collaborating with data scientists and engineers for AI tool development. The authors review definitions of bias in AI, describe common sources of bias, and present recommendations to guide quality control measures to mitigate the impact of bias in imaging AI. Understanding the terms featured in this article will enable a proactive approach to identifying and mitigating bias in imaging AI.

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Abbreviations: AI = artificial intelligence, DICOM = Digital Imaging and Communications in Medicine, ML = machine learning

TEACHING POINTS

- Understanding bias in AI requires awareness of coexisting definitions of the term *bias* framed within the context of AI development and deployment.
- Algorithm fairness is a growing area of research in ML aimed at minimizing differences in model outcomes and potential discrimination involving protected groups, as defined by shared sensitive attributes (eg, age, race, sex).
- Data distribution shift should be anticipated after clinical AI deployment, and practices must be proactive in monitoring AI to prevent clinical action based on erroneous AI results owing to data shift.
- Implementing a formal governance structure to supervise model performance can aid efforts for prospective detection of AI bias.
- Attempting to generalize models developed on specific populations to other groups, especially in the setting of known training dataset bias or discriminatory predictions, introduces inequitable bias and risks augmentation of health disparities.

Introduction

Understanding bias in artificial intelligence (AI) requires awareness of coexisting definitions of the term *bias* framed within the context of AI development and deployment. Without specific context, the term *bias* can refer to unequal preference to a person or group owing to preexisting attitudes or beliefs (1). The terms *explicit* and *implicit* distinguish intentional from unintentional bias, respectively.

Cognitive bias can be defined as systematic deviation from objective judgment (2). Intuitive thinking relying on heuristics, or mental shortcuts to simplify information processing and decision making, can be prone to cognitive bias and lead to diagnostic errors (3). Cognitive biases incorporated in model development can perpetuate existing health inequities. Furthermore, social bias may originate from a lack of diversity in data used to train a model or assumptions made during the model development process, amplifying existing social and cultural bias.

Statistical bias refers to differences between true and expected values when estimating a certain statistical quantity (ie, a model with output unrepresentative of reality) and is a common manifestation of bias in AI model development (2). Statistical bias can be framed as systematic error (bias) deriving from flaws in methodology that create reproducible differences of fixed consistent magnitude. In contrast, random error refers to unpredictable changes of varying magnitude due to chance (eg, noise). Figure 1 illustrates the differences between systematic and random error.

Systematic error should be a consideration when deploying a model on a patient population with demographics different from those of the training dataset. For example, given

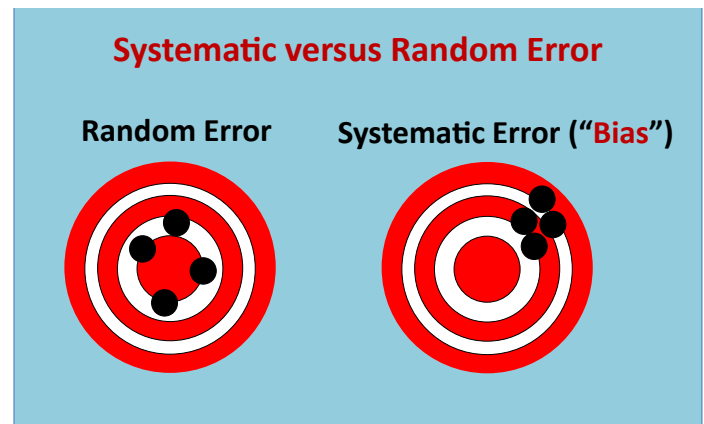


Figure 1. Diagram shows the difference between random and systematic error. Random error features unpredictable change with varying differences between each subsequent measurement or prediction. Random error primarily affects precision. Note how the black dots differ in location with varying degrees of separation from each other and the center (bull's-eye). Systematic error (bias) refers to predictable changes, with each measurement or prediction different from the previous one by a reproducible magnitude. Systematic error primarily affects accuracy. Note how the black dots differ in location but cluster with a similar degree of separation from each other and from the bull's-eye.

studies that demonstrate increased breast density in Black versus White patients, a model tasked with detecting suspicious findings on screening mammograms trained on images in White patients may consistently demonstrate decreased sensitivity if applied to mammograms in Black patients (4). Model output, in this case misclassification, would be predictable on mammograms in Black patients, following a clear pattern and demonstrating decreased accuracy despite high precision in its output in this demographic. The model would struggle to detect findings in individuals from racial and ethnic minority groups, leading to unequal rates of missed or delayed diagnosis with potentially worse clinical outcomes for non-White patients. Ensuring diverse representation in training datasets or judicious application of models based on awareness of underrepresented demographics in the training dataset will mitigate systematic error.

Random error can be seen with models tasked with automating measurements or image segmentation. Varying image quality, noise inherent in pixel data, or inconsistencies inherent to the model can contribute to random error. For example, a model tasked with measuring abdominal aortic diameter on abdomen and pelvis CT images may vary from one image to the next due to noise or artifact that leads to misclassification of pixels associated with the abdominal aortic wall. Similarly, a model tasked with segmenting the ovaries on abdomen and pelvis CT images may demonstrate random error due to inconsistent difficulty distinguishing the ovaries from other adnexal structures of similar attenuation. These models would vary in their output without following a distinct reproducible pattern of inaccurate measurement or segmentation, demonstrating decreased precision.

Prior discussions on understanding bias in AI vary in technical detail, presenting advanced content that may be

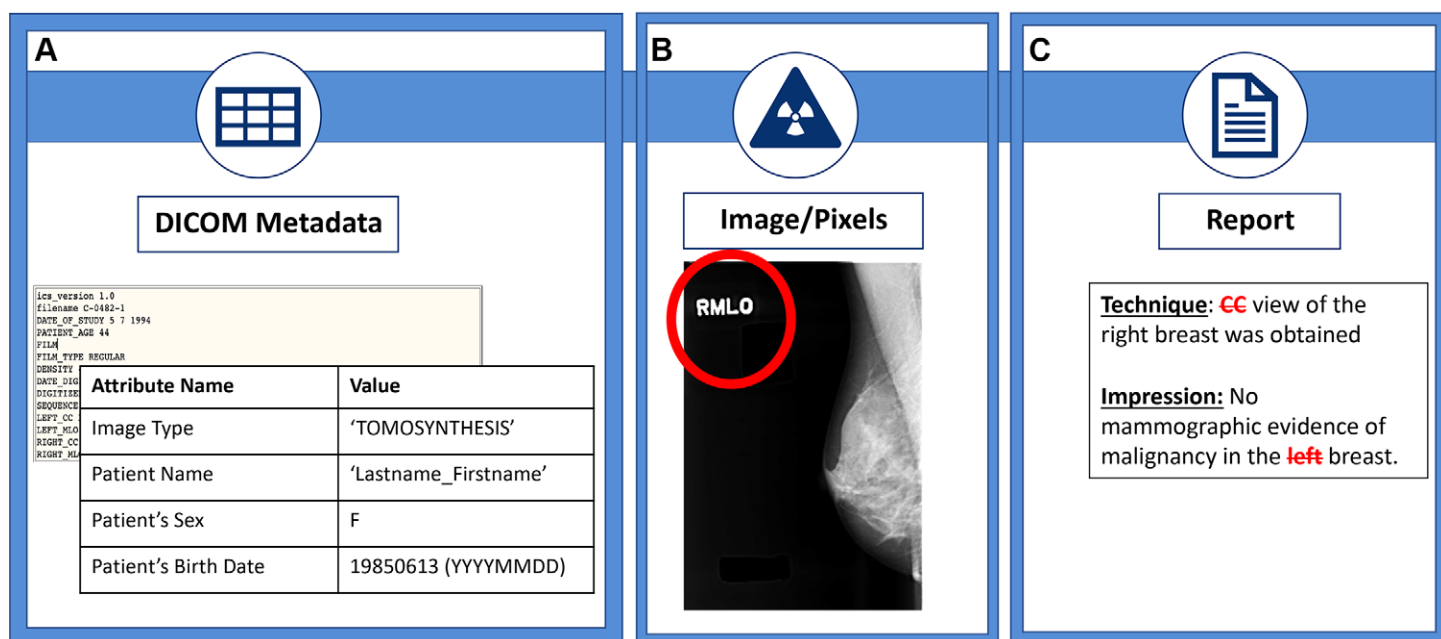


Figure 2. Diagram shows potential sources of bias in imaging AI including and beyond pixel data. Discrepancies in the Digital Imaging and Communications in Medicine (DICOM) metadata can contribute to downstream statistical bias. The left panel (**A**) shows an error in the “image type,” incorrectly attributing this examination as “tomosynthesis.” Pixel data beyond the structures of interest can also contribute to bias. The middle panel (**B**) shows a laterality marker (circle) (which is in this case correct) not removed during preprocessing of this right mediolateral (RMLLO) mammogram of the breast. Errors in the radiology report can introduce sources of bias, particularly when developing models that integrate both image and text-based data. The right panel (**C**) shows errors in the radiology report, incorrectly describing the image as a “CC” (craniocaudal) view of the left breast. (Mammogram in Fig 2B reprinted from references 11 and 12.)

intimidating for those without a technical background (5–9). However, continued growth of AI tools and clinical deployment requires a baseline knowledge of the potential for machine learning (ML) models to contribute to and perpetuate bias. Accordingly, an intuitive approach to understanding imaging AI bias involves investigating sources of bias based on stages in the ML life cycle: (a) data handling (all steps involved in data preparation including curation, investigation, annotation, splitting, and feature engineering [5,8,10]); (b) model development (process of training a model using mathematical algorithms to process, predict, and classify medical data [6,8]); (c) model evaluation (evaluating model performance per task-specific metrics involving determination of a model's ability to make predictions and generalization beyond training data [7,8]); and (d) model deployment (implementation of developed models in production environments, allowing them to make predictions on new data, including integration of AI results in daily workflow [8]).

This article focuses on the most common sources of bias that radiologists will encounter and discusses how ML models can perpetuate existing health inequities, aiming to provide a foundation for those without a technical background and actionable steps to mitigate identified sources in practice.

Sources of Imaging AI Bias

Data Handling

Pixel and Nonpixel Data.—Identifying sources of bias in AI requires evaluation beyond pixel data in medical imaging,

as seen in Figure 2, which shows an example using a mediolateral oblique mammogram from the Digital Database for Screening Mammography (DDSM) (11,12). Erroneous radiology reports or Digital Imaging and Communications in Medicine (DICOM) metadata can perpetuate bias in downstream model performance, especially for models incorporating text-based data.

For example, while a model incorporating DICOM metadata (eg, patient demographic data or image acquisition data) as labels to pretrain an algorithm may improve its ability to find patterns in a dataset, the ultimate model will reinforce bias inherent in the DICOM metadata (ie, from consistent erroneous metadata) (13). For example, consistent training of a model tasked with recommending management based on dual-energy x-ray absorptiometry findings with inaccurate age and/or sex metadata can lead to consistent underperformance due to incorrect associations between age, sex, and incidence of osteopenia or osteoporosis based on dual-energy x-ray absorptiometry findings.

While several potential sources of bias arising from pixel data are evident at image review, such as incorrect laterality labels or tissue attenuation, other confounding variables are not as clear. Researchers have shown that imaging AI models can predict demographic information, such as racial identity, from radiographs without clear explainability, demonstrating that latent associations are difficult to detect, potentially exacerbating associated clinical disparities (14). Accordingly, AI models in deployment may rely on latent associations, such as the distribution of race in a training dataset despite data de-identification, potentially exacerbating associated clinical disparities.

Table 1: Examples Illustrating the Label Ambiguity Problem

Annotator ID	Annotated Raw Text Data
Annotator 1	[Obstructing large ureteral calculi] result in [severe bilateral hydronephrosis].
Annotator 2	Obstructing large ureteral [calculi] result in severe bilateral [hydronephrosis].
Annotator 3	Obstructing large [ureteral calculi] result in severe [bilateral hydronephrosis].

Note.—Examples illustrating the label ambiguity problem (ie, multiple labels for a specified annotation task). Each line represents alternative sample label assignment by separate data annotators tasked with identifying calculi and hydronephrosis. The text in brackets identifies labeled values in each sentence. Model training on each of these inconsistently labeled sentences will influence the performance of the final model.

Dataset Source Heterogeneity.—Dataset handling necessitates sampling and organizing real-world data to create training, validation, and test datasets for AI model development. Datasets representative of a target population can ensure creation of a generalizable AI model. In practice, a minority of studies feature external validation, or assessing performance of a model on datasets derived from sources sufficiently different or geographically distant from those used for training (15). Training models with data from a single source lends to solutions that do not generalize to populations with diverse patient demographics and varying socioeconomic contexts and to studies performed on equipment from different manufacturers. Furthermore, selection of data based on convenience or availability enables sampling bias, a subset of selection bias, as training datasets subsequently do not include data that is representative of the target population.

Class Imbalance.—Limited availability of data can lead to uneven data distribution within or between training, validation, and test sets with disproportionate class representation, termed *class imbalance*. Class imbalance can occur when developing models for rare diseases, for which training data often feature an overwhelming number of negative or normal examples. Negative set bias, or overrepresentation of positive or “interesting” imaging studies, is another common cause of class imbalance (16). For example, curating a brain MRI dataset from a tertiary care center may result in overwhelming representation of MRI studies with gliomas owing to referral of patients with known disease to the cancer center for state-of-the-art management.

Sampling bias may directly drive class imbalance by contributing to minority bias due to model training on a dataset featuring undersampled protected groups (17). For example, training datasets can feature unequal racial or socioeconomic class representation based on sampling from tertiary care (private imaging centers vs county safety-net hospitals serving inherently different patient populations). Underrepresentation of classes in a training dataset forces a model to learn patterns with just a few examples before making predictions

on target data (18). Models may then assume nonexistence of classes that are not represented in the training data (19).

Data Labeling.—Labeling data for model training is often done manually and requires multiple expert human annotators that may have varying levels of expertise and prior experience. Annotator bias refers to the tendency for data annotators to project their own experience, fund of knowledge, subjective interpretation of a labeling task, and cognitive biases to a given data labeling task (10). Annotators are prone to the same cognitive biases faced by radiologists interpreting studies in clinical practice (3). For example, annotators may assign labels based on recently reviewed cases (availability bias), fail to notice findings of interest due to atypical manifestation and/or unexpected location (inattention bias), or refuse to adjust an initial label assignment despite contrary information and/or findings (anchoring bias) (3). The resulting labels can contribute to intra- and interannotator variability despite provision of standard ground truth annotations (10).

Developers may rely on external third-party annotating services for large datasets, further introducing heterogeneity that contributes to disagreements between final annotations, especially in the absence of standard labeling guidelines and expertise. Differing levels of English proficiency may introduce confusion in standard annotation guidelines, leading to inconsistent data labeling. Ultimately, these factors heighten the risk for multiple conflicting labels, termed *label ambiguity* (19). Table 1 provides an example of the label ambiguity problem using text data, which may be exacerbated in the context of multiple annotators and label schemas (20).

Discrepancies in data labeling can be amplified by automated processes for labeling large datasets using natural language processing tools that sort through image reports that, although efficient, are subject to separate sources of bias (21). While varying inconsistencies in annotation can result in noise, persistent errors in labeling due to language barriers, misinterpretation of annotation guidelines, or cognitive bias will contribute to systematic error that ultimately bias downstream model performance. For example, annotators tasked to “identify the position of the nasogastric tube” on a chest radiograph may interpret this as either denoting the distal tip of the nasogastric tube or the position of the side hole. If the ultimate model is tasked with specifically identifying the position of the side hole, misinterpretation of annotation instructions will lead to consistent inaccurate annotation of the distal tip, resulting in systematic error. Alternatively, labels from annotators asked to “identify the side hole of the nasogastric tube” will only be prone to random error owing to occasional inaccurate annotation of the side hole without a clear reproducible pattern (ie, due to difficult visualization from overlying structures).

Bias at this stage may also arise directly from labels or imaging data as opposed to interpretation of labeling instructions. Label bias occurs when options for labels in a dataset fail to capture the full range of possible labels representative of the target population, closely associated with preceding sampling bias (17).

Caution is also warranted when obtaining publicly available datasets due to a risk for unknown labeling schemas or

annotator detail, predisposing to the risks of bias discussed in this section. For example, a prior study involving a chest radiograph classification use case found inconsistent class definition, class overlap (eg, infiltrate and consolidation), and separation of findings versus diagnosis (eg, consolidation versus pneumonia) between publicly available datasets and third-party vendor labeling schemas (20). Potential mitigating factors include requesting labeling schemas for specific public datasets (schema datasheet) and defining use cases with corresponding standard schemas (20).

Model Development

Data Leakage.—Before model development, curated data must be separated into training, test, and optionally additional validation sets in a process termed *data splitting* (5). Data splitting can occur at the image, series, study, or patient level. Failure to prevent overlap between these datasets can lead to a false inflated performance on the validation and/or test datasets and poor generalization to new unseen target data.

Data leakage describes the inclusion of information in a training dataset that would not be available at the time of real-world prediction, such as overlapping training and test data. Training set contamination is a subset of data leakage that specifically refers to overlap between training and validation or test datasets. Data duplication represents a common cause of training set contamination, as redundant data may “leak” from a training dataset to validation or test datasets. Consequently, model performance may be falsely inflated during model evaluation due to training the model on patterns present in the given test set while masking potential inadequate model representation of patterns outside this context. Redundant data may present as multiple radiology examinations from the same patient obtained within a short time frame split into training, validation, and test datasets despite featuring the same pathologic features. Additionally, combining multiple datasets (termed *Frankenstein datasets*) can predispose to training set contamination due to data duplication as public datasets may feature the same smaller component datasets (22).

Missing data presents another opportunity for introducing data leakage. Real-world data commonly features missing values due to data corruption, failure to record data, human or measurement error, or withheld data due to patient preference (23). Using the mean or median of the global dataset (versus just the training dataset) as a placeholder presents an opportunity for data leakage as obtaining global statistics allows a model to train on characteristics shared by both the training and test datasets, lending to apparent high performance (19).

Improper Feature Engineering.—Feature engineering is the process of choosing and transforming data features (ie, measurable variable of interest) to enhance model performance. This process involves extracting meaningful information and eliminating irrelevant or redundant ones to improve the model’s ability to understand patterns for accurate predictions. Examples of predictive features for radiology use cases include pixel density or lesion shape to identify suspicious

findings on a mammogram, average Hounsfield units to inform prediction of intracranial hemorrhage, or patient arrival timestamps to predict examination wait time (24).

Improper feature engineering can lead to biased model development by introducing systematic errors. For example, improper cropping of images may lead to consistent over- or underrepresentation of certain features in a training dataset, skewing predictions. Consider the implications on performance for a model tasked with classifying findings on mammograms trained on data that features cropped images of easily identifiable asymmetries or masses while failing to include areas of higher breast density or regions on the margin of the image (eg, axilla).

Model Evaluation

Selecting Appropriate Performance Metrics.—Inappropriate validation metrics can misrepresent model performance and limit ultimate clinical effectiveness owing to misleading conclusions. An understanding of metric suitability for specific problem categories (eg, object detection, segmentation, classification) is crucial (25). For example, while accuracy describes model performance across all classes, its dependence on disease prevalence may provide misleading information in the context of class imbalance. Alternatively, the F1 score, the harmonic mean of precision ($[\text{true positives}] / [\text{true positives} + \text{false positives}]$) and recall ($[\text{true positives}] / [\text{true positives} + \text{false negatives}]$), is a more suitable metric for model performance in the setting of class imbalance (26,27).

In statistics, precision equates to positive predictive value and recall equates to sensitivity. For example, although a study assessing the ability of ML models to predict the development of depression in patients with knee osteoarthritis demonstrated high model accuracy on both internal and external datasets, the model demonstrated corresponding low F1 scores attributed to the presence of class imbalance in both datasets (9% and 11% of patients in the internal and external cohort diagnosed with depression, respectively) (28). Similarly, while sensitivity, specificity, and the area under the receiver operator curve may be well suited for evaluating a model’s ability to discriminate between classes, the area under the receiver operator curve would be a better choice to measure discrimination across multiple classification thresholds and analyze the trade-off between sensitivity and specificity at varying cutoffs (25).

Importance of Subgroup Analysis.—Models are often assessed based on aggregate performance based on test or evaluation datasets as opposed to determining performance on subgroups determined by factors like demographic or geographic class. However, evaluating performance in aggregate can mask potential discrepancies between subgroups, preventing identification of bias that can be exacerbated after clinical deployment in specific subgroups based on age, sex, race, or socioeconomic status. Performing subgroup analysis provides a means to identify and investigate poor performance in certain subgroups, allowing for proactive prevention of use of a given model in an ill-suited population that

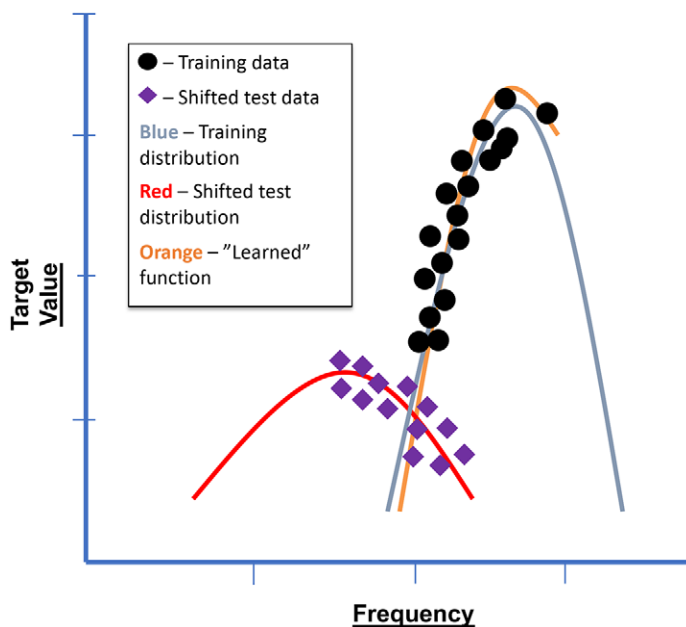


Figure 3. Graph shows covariate shift and consequent effects on ML algorithm performance. The learned function (orange) fits the training data (black circles). Differences in dataset distribution result in poor performance of the test set (purple diamonds), which is not represented by the learned function. Note that although the training and shifted test target value frequencies (red vs blue, respectively) change, the parabolic relationship between the frequency and target value is maintained for both distributions.

may be underrepresented in the model development process. Aiming for equal performance across subgroups represents a step toward ensuring model generalizability, indicating that high performance is not limited to a single subgroup (15).

Model Deployment

Data Distribution Shift.—*Data distribution shift* refers to a mismatch between training and real-world (target) data, reflecting changes in target data distribution not captured by the initial training distribution. Consequently, models demonstrating high performance in test environments experience declining performance after deployment. Developing an algorithm often involves an assumption that training and test environments are similar, which is not the case in real-world scenarios. Consequently, models may reflect bias inherent in limited training data and perform poorly on out-of-distribution data without end-user awareness.

Covariate shift, the most common type of data distribution shift, describes changes in the input distribution (range or pattern of data provided to an ML model) owing to shifting independent variables (factors driving a change in input distribution) with stable output distribution (range or pattern of predictions provided by an ML model); in this case, “stable” refers to the relationship of model input and output. Figure 3 demonstrates a maintained relationship between input feature distribution and target distribution despite changes in the former. Changes in hardware (eg, equipment from different vendors), imaging protocols, postprocessing software,

and patient demographics can introduce covariate shift. For example, one study demonstrated increased screening mammography recall rates by up to threefold following software upgrades on mammography equipment, which required adjustment to presoftware update threshold, making an argument for continuous monitoring in the setting of any software changes to avoid covariate shift (29).

Social Bias and Clinical Impact of Using Biased Models

Social Bias Can Propagate throughout Model Development

The potential for clinical decisions based on AI results to unequally impact patients is difficult to predict owing to lag time associated with a clinical decision and downstream impact. Statistical bias can perpetuate if left unchecked during model development, resulting in biased decision making that may inform the development of future model iterations. Figure 4 depicts the cyclical nature of bias in AI model development and the potential impact after deployment.

Models can make predictions on sensitive attributes, like patient race, age, sex, and ethnicity, thought to be de-identified before model development. Although recent studies demonstrate the ability for AI solutions to leverage latent information on chest radiographs to predict sensitive attributes, specific features aiding prediction of sensitive attributes remain beyond user awareness (14). While explainable AI techniques offer insight on features informing model prediction, such as the role of the sternum and upper ribs in model prediction of patient sex, identification of features consistently contributing to model prediction of sensitive attributes is still lacking (14,30,31). Accordingly, these sensitive attributes may inform ultimate model predictions, amplifying associated clinical bias inherent to a given clinical problem or dataset without end-user awareness. Similarly, models may infer demographics, baseline health factors (eg, body mass index), and manifestations of chronic illness (eg, cardiomegaly, vascular calcifications, osteopenia) from radiographs to predict cost of care (32). While an AI solution leveraging this information may inform practices to limit unnecessary health care expenditures, alternative ill-intentioned parties may use similar models acting on sensitive attributes to deny care to high-risk individuals.

Consequently, identifying a clinical problem to solve with ML requires thorough research on the context of the problem, as associated systemic bias or disparities can propagate to downstream imaging AI development. Thorough vetting of a clinical problem should include consideration of access to relevant imaging modalities by all patient groups, disparities in standards of patient referral for imaging, and differences in follow-up imaging adherence if considering assessment of treatment response. For instance, studies show (a) tendencies to more frequently order imaging in the emergency department for White versus non-White patients, (b) racial differences in follow-up rates for incidental pulmonary nodules, and (c) decreased odds for Black patients to undergo PET/CT compared with non-Hispanic White patients, potentially

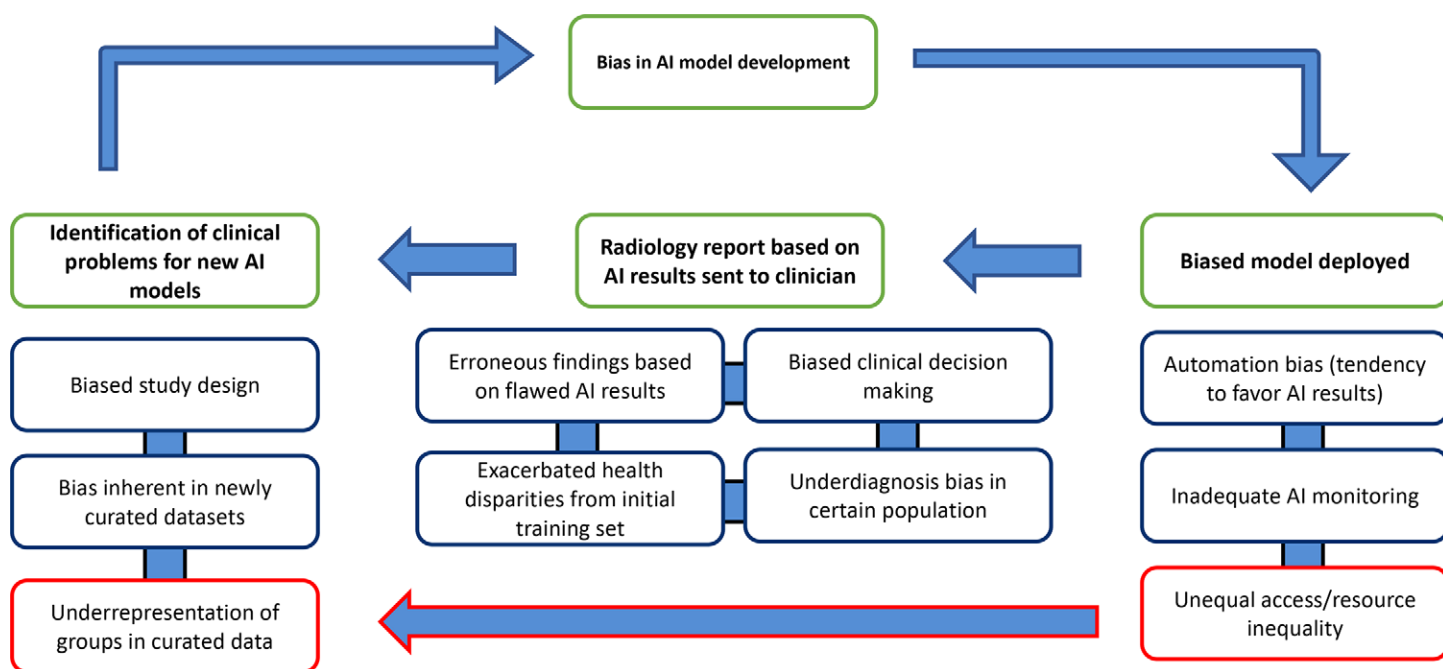


Figure 4. Flowchart shows the cyclical nature of bias in AI model development and deployment. Unchecked sources of bias during model development can result in biased clinical decision making due to errors perpetuated in radiology reports, potentially exacerbating health disparities. Identification of clinical problems and curation of data for future AI tools may then reflect bias passed forward in this cycle, perpetuating bias in development of new AI tools.

owing to lower odds of treatment at a National Cancer Institute center (33–35). These examples demonstrate existing paradigms in access to care or standards of practice that can influence AI results owing to algorithm bias originating from training datasets capturing these disparities.

How Biased Models Can Perpetuate Social Bias

Once biased models are deployed, privilege bias, or unequal access to AI solutions, may exclude certain demographics and prevent a subset of patients from receiving the same potential benefits of AI-augmented care available to others (17). Underrepresentation of certain patient demographics owing to heterogeneous access to deployed AI solutions can contribute to data shift for future iterations informed by postdeployment performance (17). Consequently, future models will be trained on biased training datasets owing to prior iterations that only represent a subset of patients, limiting conscious application of new models on underrepresented populations or risking inappropriate or inadvertent use of these models on patient populations not represented in model development.

Automation bias may enable biased models to perform unchecked, exacerbating existing social bias. Automation bias refers to favoring automated recommendations despite contrary data, effectively deferring the decision-making expertise to the automated system or machine (16,36).

Overreliance and blind agreement with AI results can result in significant interpretation errors in the absence of a deliberate approach that leverages AI results as an informative tool as opposed to a final interpretation. In clinical decision support systems, automation bias may manifest as omission versus commission errors (37). In the context of imaging AI, omission errors refer to a failure to notice or act on incor-

rect AI results that may be augmented in high-volume fast speed-reading environments. For example, in the setting of a patient presenting for a “code stroke,” missed or delayed action to alert nonradiologist clinicians (eg, neurologists) about false-positive AI results of a large vessel occlusion risks nonindicated clinical action and patient harm in the form of unnecessary thrombolytic provision on the basis of AI results directly conveyed to the clinical team (38). Commission errors refer to accepting incorrect AI results despite contrary evidence, more commonly seen in the setting of less experienced practitioners and subtle findings. In contrast, increased user confidence in a given task is associated with less reliance on automated results (37,39). Similarly, users are prone to automation bias with more complex tasks in time-constrained environments, such as due to increased workload (37).

In radiology, these factors translate to a higher risk of automation bias owing to high-volume clinical contexts demanding rapid turnaround times for reporting heterogeneously complex imaging examinations. Similar risks are compounded by lack of task experience among junior residents or faculty training in high-volume settings that use AI. For example, a study assessing the impact of radiologist experience on agreement with AI results demonstrated that relatively inexperienced radiologists reading mammograms are more likely to follow incorrect AI suggestions for a higher Breast Imaging Reporting and Data System (BI-RADS) category compared with more experienced readers (40). Accordingly, although AI solutions offer means to assist pressured radiologists in this context, efficiency gains are accompanied by a risk of unverified erroneous AI results adversely impacting patient care.

The effects of automation bias may persist beyond an immediate clinical decision due to an erroneous feedback loop,

Table 2: Fairness Metrics	
Fairness Metric	Summary
Statistical or demographic parity	Requires equal probability of positive predicted class across protected and unprotected groups
Disparate impact	Represents the ratio of the rate of positive prediction between protected and unprotected groups
Calibration	Requires similarity between probability prediction or risk scores and actual outcomes regardless of group
Predictive parity	Requires equal positive predictive values across protected and unprotected groups
Error rate	Represents the ratio of incorrect to total predictions compared among protected and unprotected groups
Equal opportunity	Requires that a preferred outcome is predicted equally across protected and unprotected groups
Equalized odds	Requires equal true-positive and false-positive rates between protected and unprotected groups

as acceptance of incorrect AI results contributes to continuous model training, prompting a similar mistake with future use (17). Imagine consistent acceptance of incorrect AI results from an algorithm tasked with detection of cervical spine fractures that fails to flag fractures in patients with bone demineralization. Positive feedback for this model (agreeing that there is no fracture when one is present) will continue to result in failure to identify acute pathologic conditions, as additional false-negative cases will reinforce model output and low sensitivity.

Moreover, automation bias carries a tendency to disproportionately affect certain patients. Lower socioeconomic patient populations or resource-constrained health systems are at higher risk of exposure to automation bias. Resource- or expert-constrained systems, particularly featuring a paucity of AI-savvy radiologists, are prone to overreliance on AI solutions owing to understaffed services with high demand and increasingly complex studies. Lack of expert radiologist review of AI results in these contexts risks downstream clinical action, or the lack thereof, based on inaccurate AI results.

Inequity and Inequality

Inequity and inequality represent two distinct noninterchangeable terms. *Inequity* refers to unjust avoidable differences in health or distribution of resources between people of different social, economic, geographic, or demographic groups, making certain groups more vulnerable to poor outcomes due to higher health risks (41,42). In other words, inequity refers to unfair access to the same opportunities as another group owing to injustice. Note that inequities are driven by social determinants of health and can be remediated with appropriate intervention. In contrast, *inequality* refers to unequal differences in health outcomes or distribution of resources without

reference to fairness (41,42). Measuring health inequalities can allow an indirect assessment of health inequity.

Prior work has shown potential for AI models to create or exacerbate biases resulting in differences in performance among certain populations and amplification of health inequities (43–49). Underdiagnosis bias, or falsely claiming absence of pathologic conditions with consequent clinical inaction and delay in care, is one potential means of AI-related exacerbation of health inequity (50). For example, one study found imaging AI models for chest radiographs were prone to underdiagnosis for female, young (<20 years of age), Black, Hispanic, and Medicaid-insured patients, potentially partially deriving from bias inherent in labels obtained from clinical records (50).

Concerns regarding amplification of health inequity by AI systems stem from a potential to capture social determinants of health or cognitive bias inherent in real-world data (51). For example, Obermeyer et al (45) demonstrate that low algorithmic risk scores used to screen patients for “high-risk care management” programs may reflect “healthy” White patients while failing to capture Black patients with relatively higher illness burden that should qualify for care management programs. Further results from that study suggest that the finding of higher illness burden for Black patients at a given risk score compared with White patients derives from the algorithm predicting health care costs as opposed to illness, reflecting unequal access to care and lower health care dollars spent on Black versus White patients (45). Other examples indicate inadvertent assignment of overbooked appointment slots to Black patients by automated scheduling systems that potentially rely on prior no-show rates influenced by social determinants of health (52). Similarly, imagine the impact of an algorithm tasked with detecting breast lesions on mammograms trained on a dataset featuring potentially missed lesions associated with socioeconomic factors including income level, literacy, and insurance status (53).

Fairness

Algorithm fairness is a growing area of research in ML aimed at minimizing differences in model outcomes and potential discrimination involving protected groups, as defined by shared sensitive attributes (eg, age, race, sex) (6,54). In this context, unfair algorithms are those that favor certain groups over others based on sensitive attributes. Prior work describes fairness metrics that may differ in reliance on predicted probabilities, predicted outcome, actual outcome, and emphasis on group versus individual fairness (54,55). Common fairness metrics are summarized in Table 2 (54–56).

Note that certain fairness metrics may conflict with others based on the specific algorithmic task and outcome rates among groups, necessitating judgment for appropriate application of each metric or definition based on task context (55,57). Selecting a single fairness metric may lead to inevitable determination and potentially unrealistic representation of algorithm unfairness.

Recommendations to Mitigate Bias

Prior articles propose strategies for mitigating statistical bias at data handling, model development, and model evaluation

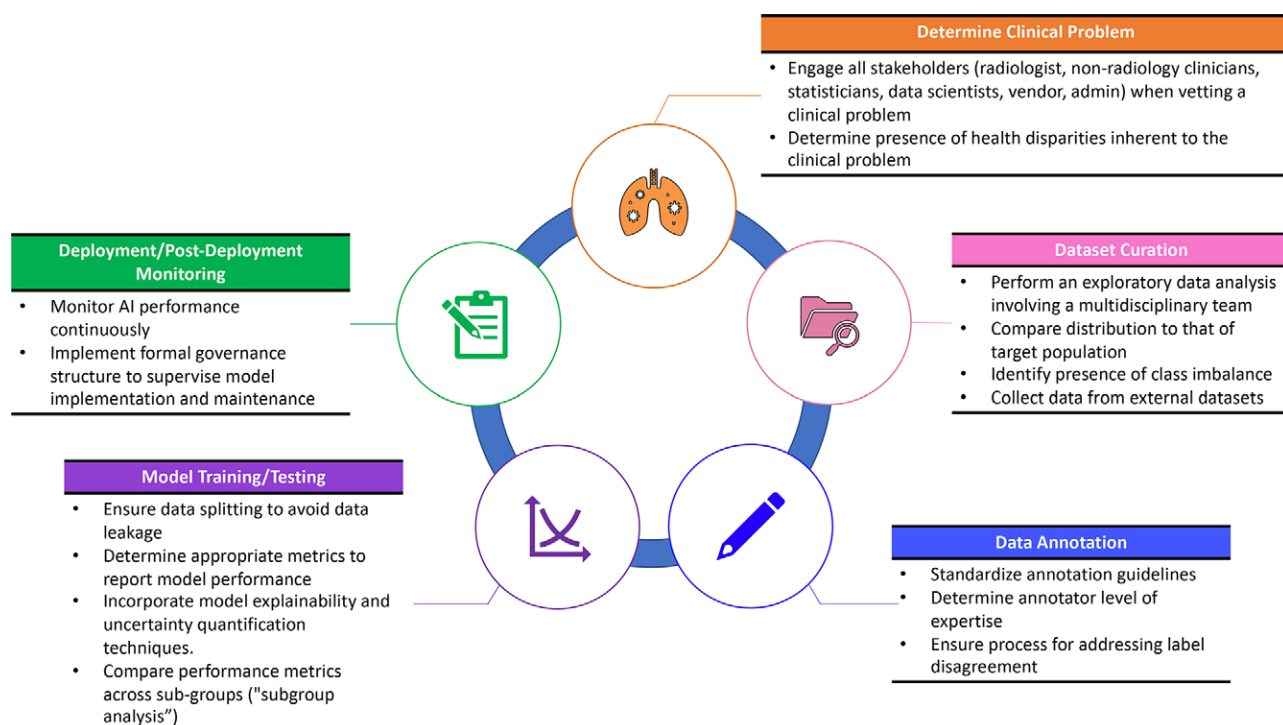


Figure 5. Proposed recommendations addressing sources of bias during AI development and after deployment. A strategic approach to mitigating bias focuses on adopting checkpoints along the ML life cycle to prospectively address potential sources of bias.

levels (5–7). Additionally, frameworks have been suggested to address potential discriminatory actions prompted by biased AI results (58,59). Combining these strategies and others from nonradiology literature allows guidance for best practices to minimize the impact of bias when developing or deploying AI in clinical practice. Figure 5 details recommendations to address the sources of bias discussed in this review to establish best practices that mitigate statistical and algorithm bias in AI. A strategic approach can focus on mitigating bias at each of the described stages of the ML life cycle.

Engaging Stakeholders to Define a Clinical Problem

When deploying an AI model, start by engaging all stakeholders to thoroughly define the clinical problem and assess potential sources of downstream bias. An AI task force composed of radiologists, nonradiologist clinicians, statisticians, data scientists, informaticists, technologists, and department administrators will provide multidisciplinary perspectives to plan successful deployment, AI workflow maintenance, and long-term AI monitoring. Consider incorporating individuals from diverse backgrounds to ensure representation of all potential groups and foster meaningful discussion about the impact of existing health disparities (58). Factors that define the clinical question should be thoroughly vetted for historical evidence of health inequity, such as unequal referral practices, follow-up adherence, or disease manifestation that may be perpetuated by AI (58).

Thoughtful Dataset Curation

An interdisciplinary team incorporating domain expertise is vital for thorough data investigation after data curation (5). Data investigation should include exploratory data analysis

to organize raw data, determine patterns, and frame the data in context of the clinical problem (5). For example, evaluate sources of data used to train the algorithm, including determining the number of sites of data curation, and assess characteristics of aggregated sources (eg, large public datasets composed of subdatasets).

Encountering missing data is not uncommon during this process due to human error, equipment malfunction, or patient decision to not disclose information (23). Deletion and imputation represent common approaches to address missing values. Deletion involves removing rows or columns with missing values in a given dataset. Imputation involves replacing missing values with a placeholder or substitution value. Figure 6 demonstrates common methods for deletion and imputation. However, opting to delete input categories with missing values risks worsening model performance due to a relative paucity of data or exacerbation of inherent bias owing to class imbalance.

Mitigating the risk of unfair models perpetuating health disparities begins with prospective evaluation of dataset composition after data collection. Curated data should be analyzed for demographic, socioeconomic, geographic, and genotypic characteristics to determine appropriate representation of the intended patient population for ultimate AI deployment.

Model Training and Testing

Ensuring splitting of training, validation, and test datasets is the most common means of preventing data leakage. However, splitting should be ensured at the patient level to avoid unintended training set contamination (5). Splitting data at the image, series, or study level still risks data leakage due to potential shared features among adjacent images, reconstructed

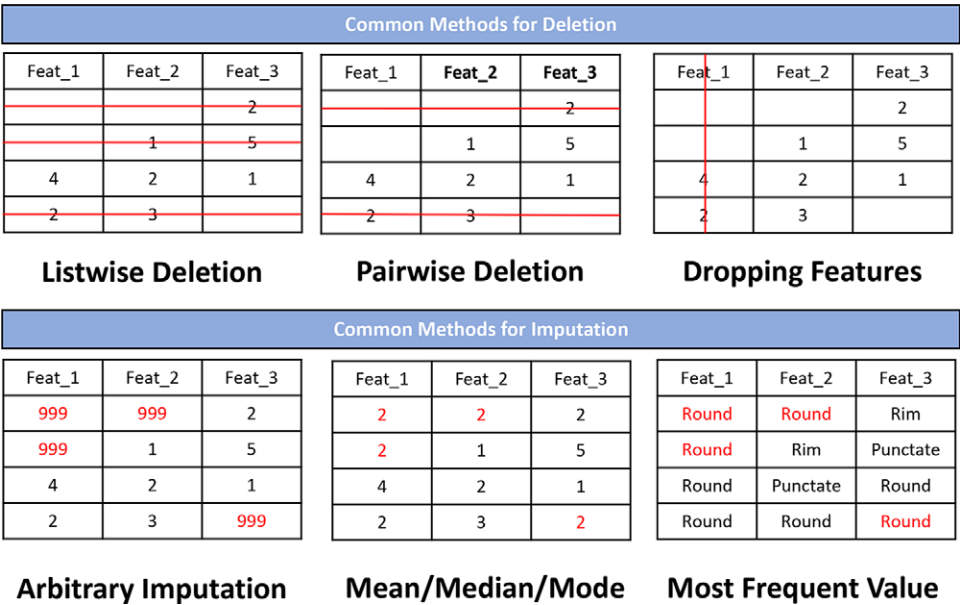


Figure 6. Charts show common methods of deletion (red lines) and imputation (red numbers and text) to address missing data. Deletion involves removing rows or columns with missing data. Listwise deletion removes any row with missing values. Pairwise deletion removes rows with missing values only in columns of interest (eg, if performing analysis for bolded Feat_2 and Feat_3). Dropping features involves removing columns with missing values under a defined threshold (eg, more than 25% values are missing). Imputation replaces missing values with a placeholder or substitution value. Arbitrary imputation replaces missing values with arbitrary values (eg, 0, 999, -999). Missing values may be replaced with the mean, median, or mode of the data. Most frequent values replaces missing categorical data with the most common category.

series of the same pathologic features, or redemonstration of pathologic features at subsequent studies.

During testing, incorporate data scientists and statisticians to determine appropriate performance metrics to assess model performance. Consider supplementing this process with open-source tools summarizing recommendations for image analysis validation, such as the Metrics Reloaded online tool (25).

Before deployment, model performance should be evaluated in aggregate as well as in subgroups to uncover potential discrepancies between protected and nonprotected groups, potentially masked by aggregate model performance.

Model Deployment and Postdeployment Monitoring
Data distribution shift should be anticipated after clinical AI deployment, and practices must be proactive in monitoring AI to prevent clinical action based on erroneous AI results owing to data shift. Continuous AI monitoring, specifically model predictions, allows identification of degrading model performance and associated factors, such as inclusion of new clinical sites with different patient demographics or introduction of new equipment. Although dashboards can display changes in AI performance over time, collaboration with statisticians and data scientists is key to deriving actionable insights to correct model performance. Detailed monitoring may extend beyond model predictions to examining model input and associated predictive power. One corrective step may involve adjusting for specific input features driving data shift. More elaborate measures include retraining models after determining the time interval for new training data sampling (eg, weekly, monthly, annually, or from the onset of worsening model performance) (19).

Implementing a formal governance structure to supervise model performance can aid efforts for prospective detection of AI bias. Prior work presents a framework for flexible governance of clinical AI implementation and performance, incorporating fairness and bias as a metric for evaluating a model

for clinical implementation (60). Formal governance with either radiology- or enterprise-level leadership should involve routine assessment of checkpoints in place to mitigate systemic biases.

Equitable Bias
Completely eliminating sensitive attributes to satisfy algorithm fairness may counterintuitively induce harm by excluding factors that capture unequal outcome distribution in a target population (6,54,56). While inequitable biases can potentially harm patients, equitable biases may address challenges associated with underrepresentation of minority groups with consequent poor model performance (61). Using strategies such as oversampling data from underrepresented populations with the intent of deploying the model in those specific groups can address specific needs overlooked by more general models (61).

Alternatively, generative AI models can be used to create synthetic data to augment datasets featuring gaps in demographic or socioeconomic representation (62). However, caution is warranted when using generative AI models as recent state-of-the-art models have been shown to perpetuate racial and gender stereotypes, for example, by featuring synthetic images of White male individuals when prompted for “portraits of CEOs” and minority women for “portraits of social workers” or individuals belonging to racial minority groups for “forensic sketches of potential suspects” (63). Furthermore, model training on data generated by other models predisposes to model collapse, a process where models “forget” the true data distribution converging on a single point estimate with small variance over time (64).

Attempting to generalize models developed on specific populations to other groups, especially in the setting of known training dataset bias or discriminatory predictions, introduces inequitable bias and risks augmentation of health disparities (61). Models trained on specific populations are prone to overfitting, resulting in poor generalization to other contexts. While generalization of models through external validation

Table 3: Select Sources of Bias during Data Handling and Mitigating Steps

Bias	Summary	Mitigating Steps
Sampling bias or data source homogeneity	Data curated from a single source or similar geographic distribution, lacking representation of diverse patient demographics, varying socioeconomic contexts, and multiple vendor equipment or acquisition protocols	Curate data from multiple sources with varying demographic, socioeconomic, and geographic distribution Determine if source of data represents intended patient population
Class imbalance	Uneven data distribution within or between training, validation, and test sets with disproportionate class representation	Consider data augmentation techniques Oversample underrepresented classes Caution with use of generative AI due to risk of model collapse
Annotator bias	Tendency for annotators to project their experience, fund of knowledge, subjective interpretation of a labeling task, and cognitive biases to a given data labeling task	Include annotators with varying experience and backgrounds while maintaining threshold for task experience Include intra- and interrater analysis
Label bias	Options for labels in a dataset fail to capture the full range of labels representative of a target population	Involve multiple stakeholders and perspectives in review of annotation guidelines to ensure representation of full spectrum of labels
Presentation bias	Presentation of data in a way that systematically influences label assignment	Involve multiple stakeholders in design of presentation modes Include comparison of different modes of presentation

Table 4: Select Sources of Bias during Model Development and Mitigating Steps

Bias	Summary	Mitigating Steps
Data leakage	Inclusion of information in a training dataset that would not be available at the time of real-world prediction, such as training contamination (overlap between training and validation or test datasets)	Ensure data splitting without inclusion of training data in the test set Use the mean or median of the training data only to address missing training data as opposed to using the mean or median of the global dataset
Redundant data	Multiple radiologic examinations from the same patient obtained within a short time frame split into training, validation, and test datasets despite featuring the same pathologic conditions	Ensure data splitting at the patient level Consider risk of data duplication from combining multiple public datasets (Frankenstein datasets)
Missing data	Missing data due to data corruption, human error, malfunctioning equipment, or patient decision	Use deletion or imputation techniques
Improper feature engineering	Inappropriate selection and transformation of data features leading to over- or underrepresentation of certain features	Careful review to avoid over- or underrepresentation of features
Improper feature scaling	Normalizing data on different scales using global statistics before data splitting, leading to data leakage	Perform feature scaling on the training set versus the global dataset

presents an opportunity to address this issue, model performance with subgroup analysis should still be monitored to ensure that performance does not differ among racial or socioeconomic groups (15).

Conclusion

Imaging AI bias is a complex topic that depends on the context of AI development and deployment. The concepts presented in this article provide a foundation for understanding sources of bias in imaging AI development and manifestation of bias after model deployment. These concepts provide guidance for proactive steps to mitigate the downstream effect of AI bias, as summarized in Tables 3–5 (10). Although a more detailed technical discussion of AI bias is beyond the scope of this article, we hope that this serves as a consolidated ref-

erence for those creating or acquiring AI solutions to identify and address AI bias before realizing its impact in retrospect.

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Table 5: Select Sources of Bias during Model Evaluation and Deployment and Mitigating Steps

Bias	Summary	Mitigating Steps
Inappropriate performance metrics	Inappropriate validation metrics can misrepresent model performance and limit ultimate clinical effectiveness owing to misleading conclusions	Determine appropriate metrics based on the task Refer to literature and tools (eg, Metrics Reloaded) to understand appropriate use of common performance metrics
Neglecting subgroup analysis	Evaluating performance in aggregate, masking potential discrepancies between subgroups, prevents identification of bias that can be exacerbated after clinical deployment in specific subgroups based on age, sex, race, or socioeconomic status	Compare performance across subgroups, such as age, sex, race, and socioeconomic status
Data distribution shift	Mismatch between training and real-world (target) data, reflecting changes in target data distribution not captured by the initial training distribution	Monitor AI performance continuously Incorporate formal AI governance for AI monitoring Determine need to retrain model and time interval for new training data sampling
Automation bias	Favoring automated recommendations despite contrary data, effectively deferring the decision-making expertise to the automated system or machine	Educate AI users about automation bias and limitations of deployed AI tools Share model details about real-time performance and limitations, including error analysis

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